

CIS 6270

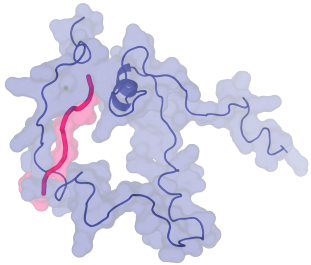
Module 5: Discrete Flow Matching

Lecture 5.4: Guidance, Rectification, and Multi-Objective Refinement

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We can keep the trained generator and use a property model to guide sampling.

Could we change the jump rates to favor sequences with the property we want?

Discrete Guidance

UNLOCKING GUIDANCE FOR DISCRETE STATE-SPACE DIFFUSION AND FLOW MODELS

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Nisonoff,..,Listgarten. *ICLR* (2025)

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With Discrete Guidance, we write the rates of the conditioned process in closed form, with no sum over sequences.

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We rebuild the diagonal as minus the row sum, so the tilted rates are still a velocity.

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That function satisfies the backward equation of the unguided process, so it is space-time harmonic.

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A partly generated sequence is useful when its possible completions are likely to satisfy the goal.

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So the guided sampler is the model conditioned on the property, exact in the continuous-time limit and with the true property model.

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For a 100-residue peptide over 20 amino acids and a mask token, that is 2001 property evaluations per step, against 21 raised to the 100 states.

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The likelihood ratios for A, C, G, and T are 0.50, 1.75, 2.75, and 1.25. Multiplying by the unguided rates gives 0.125, 0.875, 2.063, and 1.250.

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The total exit rate rises from 2.50 to 4.31, so the guided chain also unmasks faster.

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Only γ = 1 is exact because a power of h is generally not harmonic. Larger γ increases the preference for the property and samples a tempered distribution.

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The Δt factors cancel between the two expansions, so the product of rates is well defined.

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We need one forward pass and one backward pass per step, compared with 1901 forward passes for separate destination scores.

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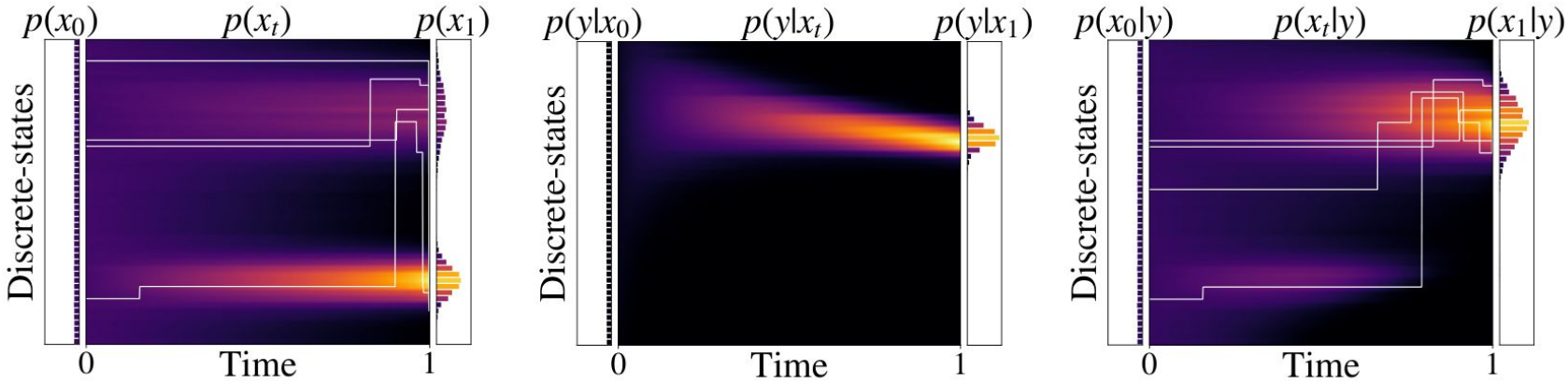
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A move becomes more frequent when it makes the desired property more likely.

With a property model on noisy states, an unconditional sampler becomes a conditional one



The unguided chain, the property model on noisy states, and the guided chain.

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Guided inverse folding on a hard protein

Method	$P_\theta(\mathbf{x} \mid \mathbf{c})$ Temp	$\Delta\Delta G > 0$ % (↑)	RMSD ≤ 2Å % (↑)	Diversity (↑)	Success (↑)
ProteinMPNN	1.0	0%	47%	53.07	0%
	0.1	1%	100%	27.98	1%
	0.01	0%	100%	27.42	0%
FMIF	1.0	0%	75%	50.29	0%
	0.1	2%	100%	23.13	2%
	0.01	4%	100%	21.82	4%
DiGress ($\gamma = 1$)	0.1	0%	100%	18.65	0%
DiGress ($\gamma = 10$)	0.1	0%	100%	16.76	0%
DiGress ($\gamma = 100$)	0.1	0%	100%	22.54	0%
FMIF-DG-TAG ($\gamma = 1$)	0.1	9%	100%	24.15	9%
FMIF-DG-TAG ($\gamma = 10$)	0.1	52%	100%	23.26	52%
FMIF-DG-TAG ($\gamma = 100$)	0.1	94%	100%	20.82	94%
FMIF-DG-Exact ($\gamma = 1$)	0.1	10%	100%	24.66	10%
FMIF-DG-Exact ($\gamma = 10$)	0.1	58%	100%	25.78	58%
FMIF-DG-Exact ($\gamma = 100$)	0.1	100%	100%	19.72	100%

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Among ProteinMPNN sequences that fold, 1% are predicted to be stable

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The unguided flow model reaches 4% at its best temperature

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The Taylor approximation with γ = 100 reaches 94%

Method	$P_\theta(\mathbf{x} \mid \mathbf{c})$ Temp	$\Delta\Delta G > 0$ % (↑)	RMSD $\leq$ 2Å % (↑)	Diversity (↑)	Success (↑)
ProteinMPNN	1.0	0%	47%	53.07	0%
	0.1	1%	100%	27.98	1%
	0.01	0%	100%	27.42	0%
FMIF	1.0	0%	75%	50.29	0%
	0.1	2%	100%	23.13	2%
	0.01	4%	100%	21.82	4%
DiGress ($\gamma = 1$)	0.1	0%	100%	18.65	0%
DiGress ($\gamma = 10$)	0.1	0%	100%	16.76	0%
DiGress ($\gamma = 100$)	0.1	0%	100%	22.54	0%
FMIF-DG-TAG ($\gamma = 1$)	0.1	9%	100%	24.15	9%
FMIF-DG-TAG ($\gamma = 10$)	0.1	52%	100%	23.26	52%
FMIF-DG-TAG ($\gamma = 100$)	0.1	94%	100%	20.82	94%
FMIF-DG-Exact ($\gamma = 1$)	0.1	10%	100%	24.66	10%
FMIF-DG-Exact ($\gamma = 10$)	0.1	58%	100%	25.78	58%
FMIF-DG-Exact ($\gamma = 100$)	0.1	100%	100%	19.72	100%

Nisonoff,..,Listgarten. *ICLR* (2025)

Exact guidance reaches 100%, with every sequence folding to within 2 angstroms

Method	$P_\theta(\mathbf{x} \mid \mathbf{c})$ Temp	$\Delta\Delta G > 0$ % (↑)	RMSD ≤ 2Å % (↑)	Diversity (↑)	Success (↑)
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DiGress ($\gamma = 1$)	0.1	0%	100%	18.65	0%
DiGress ($\gamma = 10$)	0.1	0%	100%	16.76	0%
DiGress ($\gamma = 100$)	0.1	0%	100%	22.54	0%
FMIF-DG-TAG ($\gamma = 1$)	0.1	9%	100%	24.15	9%
FMIF-DG-TAG ($\gamma = 10$)	0.1	52%	100%	23.26	52%
FMIF-DG-TAG ($\gamma = 100$)	0.1	94%	100%	20.82	94%
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Nisonoff,..,Listgarten. *ICLR* (2025)

On another protein the same γ = 100 collapses to 42%, so the strength is a hyperparameter

Method	$P_\theta(\mathbf{x} \mid \mathbf{c})$ Temp	$\Delta\Delta G > 0$ % (↑)	RMSD $\leq$ 2Å % (↑)	Diversity (↑)	Success (↑)
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Nisonoff,..,Listgarten. *ICLR* (2025)

What if we need to improve several peptide properties at once?

MOG-DFM: Multi-Objective-Guided Discrete Flow Matching

Multi-Objective-Guided Discrete Flow Matching for Controllable Biological Sequence Design

Tong Chen[1,2], **Yinuo Zhang**[1,3], **Sophia Tang**[1,4], **Pranam Chatterjee**[1,5,6,†]

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[5]Department of Computer Science, Duke University
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The goal is the Pareto front:

$$\mathrm{PF} = \{\, x \;:\; \nexists\, x' \text{ with } s_n(x') \geq s_n(x)\ \forall n \text{ and } s_m(x') > s_m(x) \text{ for some } m \,\}$$

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In MOG-DFM, we steer toward one point on that front, chosen by a weight vector ω drawn from a simplex lattice.

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First by rank, within objective n: $$I_n(y^i, x) = \frac{\mathrm{rank}\big(s_n(x^{(i \to y^i)}) - s_n(x)\big)}{|\mathcal{T}|}$$

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Second by direction, along the chosen trade-off: $$D(y^i, x, \omega) = \Delta \mathbf{s}(y^i, x) \cdot \omega$$

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We use ranks to compare objectives with different units. The inner product measures progress along the chosen direction.

The two scores become an exponential tilt

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$$\Delta S(y^i, x, \omega) = \mathrm{Norm}\left[\frac{1}{N}\sum_n i_n I_n(y^i, x)\right] + \lambda\,\mathrm{Norm}\left[D(y^i, x, \omega)\right]$$

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$$u^i_{\mathrm{guided},t}(y^i, x \mid \omega) = \beta\, u^i_t(y^i, x)\, \exp\big(\Delta S(y^i, x, \omega)\big)$$

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We use a heuristic tilt here. Without an h(y)/h(x) ratio, we cannot condition exactly.

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A good score speeds up a move only if the base model already permits it.

Accept a candidate only if its improvement sits inside a cone

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Accept a candidate only if its improvement sits inside a cone around ω:

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The alignment angle is the arccosine of the normalized inner product between the score change and ω. We accept the candidate when that angle is at most Φ.

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Inside the cone, accepted moves improve the weighted objective in expectation because their directional scores are positive. The theorem requires Φ to be less than a right angle.

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We widen the cone when the rejection rate is too high and narrow it when the rate is too low. This is proportional control of log Φ.

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Removing the cone costs most of the half-life gain in their ablation, from about 30 hours down to about 12.

MOG-DFM: which improvement fits the trade-off?

Toy improvements on two equally weighted objectives are (2, 1) for edit G and (–1, 2) for edit C. The chosen direction is (0.5, 0.5), with a 30-degree acceptance cone.

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$$\alpha_{\mathrm{G}} = \arccos \frac{3}{\sqrt{10}} \approx 18.4^{\circ}, \qquad \alpha_{\mathrm{C}} = \arccos \frac{1}{\sqrt{10}} \approx 71.6^{\circ}$$

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We use the cone to retain improvements that follow the requested balance between objectives.

Score every candidate, keep the ones inside the cone, and take one jump

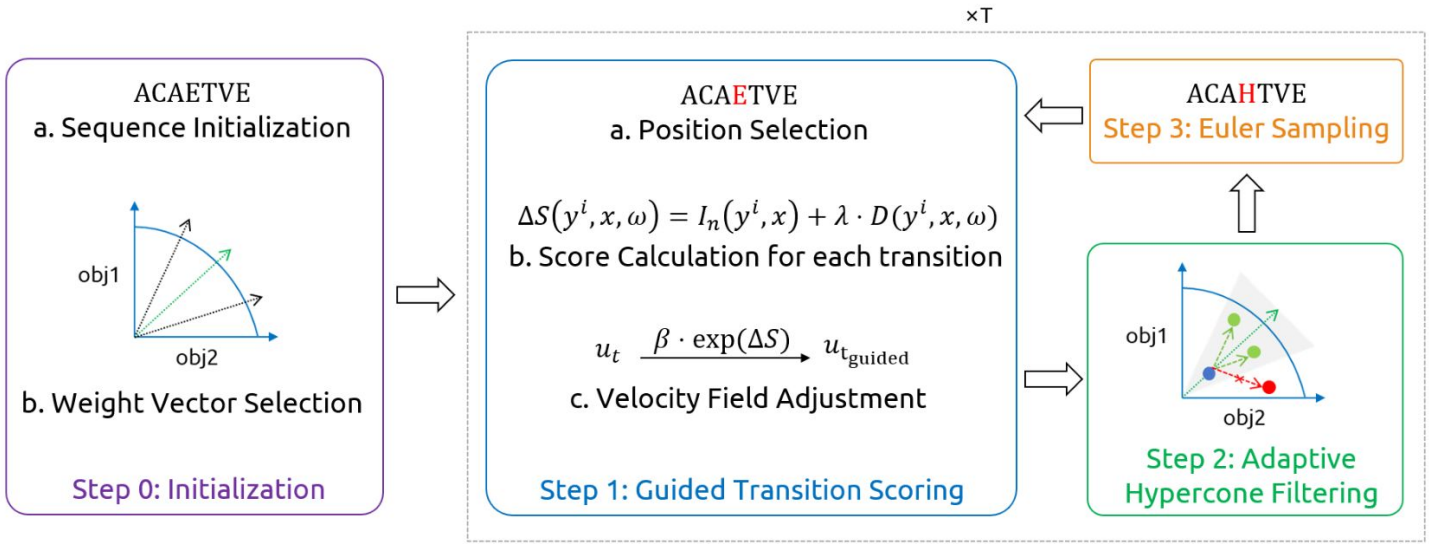


Figure 1: Visualization for MOG-DFM algorithm.

We draw one weight vector, then repeat scoring, filtering, and sampling.

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Let’s compare with classical multi-objective optimizers

Target	Method	Time (s)	Hemolysis (↓)	Non-Fouling	Solubility	Half-Life	Affinity
1B8Q	MOPSO	8.54	0.1066	0.4763	0.4684	4.449	6.0594
	NSGA-III	33.13	0.0862	0.5715	0.5825	7.324	7.2178
	SMS-EMOA	**8.21**	0.1196	0.3450	0.3511	3.023	5.955
	SPEA2	17.48	0.0819	0.4973	0.5057	4.126	**7.324**
	MOG-DFM	43.00	**0.0785**	**0.8445**	**0.8455**	**27.227**	5.9094
PPP5	MOPSO	11.34	0.0883	0.4711	0.4255	1.769	6.6958
	NSGA-III	37.30	**0.0479**	0.7138	0.7066	2.901	7.3789
	SMS-EMOA	**8.43**	0.1242	0.4269	0.4334	1.031	6.2854
	SPEA2	19.02	0.0555	0.6221	0.6098	2.613	**7.6253**
	MOG-DFM	90.00	0.0617	**0.7738**	**0.751**	**27.775**	6.8197

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A genetic algorithm reaches non-fouling 0.57 and a half-life of 7.3 hours on this target

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MOG-DFM reaches 0.84 and 27.2 hours, with hemolysis also lower

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Binding affinity is 5.91, compared with 7.32 for the best baseline, showing the trade-off with the other properties

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We also improve non-fouling, solubility, and half-life on the second target, using two to ten times the baseline runtime

Target	Method	Time (s)	Hemolysis (↓)	Non-Fouling	Solubility	Half-Life	Affinity
1B8Q	MOPSO	8.54	0.1066	0.4763	0.4684	4.449	6.0594
	NSGA-III	33.13	0.0862	0.5715	0.5825	7.324	7.2178
	SMS-EMOA	**8.21**	0.1196	0.3450	0.3511	3.023	5.955
	SPEA2	17.48	0.0819	0.4973	0.5057	4.126	**7.324**
	MOG-DFM	43.00	**0.0785**	**0.8445**	**0.8455**	**27.227**	5.9094
PPP5	MOPSO	11.34	0.0883	0.4711	0.4255	1.769	6.6958
	NSGA-III	37.30	**0.0479**	0.7138	0.7066	2.901	7.3789
	SMS-EMOA	**8.43**	0.1242	0.4269	0.4334	1.031	6.2854
	SPEA2	19.02	0.0555	0.6221	0.6098	2.613	**7.6253**
	MOG-DFM	90.00	0.0617	**0.7738**	**0.751**	**27.775**	6.8197

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Could we guide insertions and deletions while also enforcing constraints on the finished sequence?

pCoMole: Pareto-Constrained Molecule Editing

pCoMole: Pareto-Constrained Molecule Editing with Discrete Flows

Tong Chen,[1,*] **Maximilian Holsman,**[2,*] **Lin Zhao,**[3,*] **Pranam Chatterjee**[1,3,†]

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[2]Department of Computer Science, Duke University
[3]Department of Bioengineering, University of Pennsylvania

[*]Equal contribution
[†]**Corresponding author:** pranam@upenn.edu
Code: https://huggingface.co/ChatterjeeLab/pCoMole

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Take the same question to Edit Flows

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Edit Flows change the length, so the same multi-objective question arises for sequences that grow and shrink.

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Score a sequence by its worst objective, plus a small sum:

$$U(x) = \min_k \omega_k\big(f_k(x) - r_k\big) \;+\; \rho \sum_k \omega_k\big(f_k(x) - r_k\big)$$

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The utility of a sequence is the smallest weighted gap between an objective value $f_k(x)$ and its reference level r_k, plus ρ times the sum of those weighted gaps.

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The first term rewards the weakest objective, so a sequence cannot trade solubility for affinity. The second term breaks ties.

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We assign zero reward to infeasible sequences, regardless of their other scores.

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With the inverse temperature β, we set how strongly high utility is weighted.

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The terminal reward is the exponential of β times the utility, multiplied by the indicator that the sequence is feasible.

We assign zero reward to infeasible sequences, regardless of their other scores.

A high property score only matters after the finished sequence passes every feasibility check.

Put that reward through the same h-transform

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We recover the Discrete Guidance construction with a reward in place of an event. Along the path, h is a martingale whose terminal value reweights the endpoint distribution.

We estimate that expectation by rolling the flow forward

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From each candidate, run R short continuations to t = 1:

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We generate rollouts with the base Edit Flow, so we do not need to retrain it.

We assign zero value to candidates whose rollouts are all infeasible. If every candidate has zero value, we keep the current sequence.

Choose one of the candidates at each step

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$$\boxed{\tilde{u}_t(y_c \mid x_t) \;\propto\; u_t(y_c \mid x_t)\,\hat{h}_t(y_c), \qquad c = 1, \ldots, C}$$

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This is a selection rule over sampled candidates, because a variable-length state space has too many destinations for a full rate.

pCoMole: two feasible molecule edits

Toy edits P and Q have weighted objective gaps (0.4, 0.1) and (0.2, 0.2). Both are feasible and have equal base rates. Take the sum coefficient as 0.1 and reward strength as 10, with deterministic rollouts.

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$$U(P) = 0.1 + 0.1(0.4 + 0.1) = 0.15, \qquad U(Q) = 0.2 + 0.1(0.2 + 0.2) = 0.24$$

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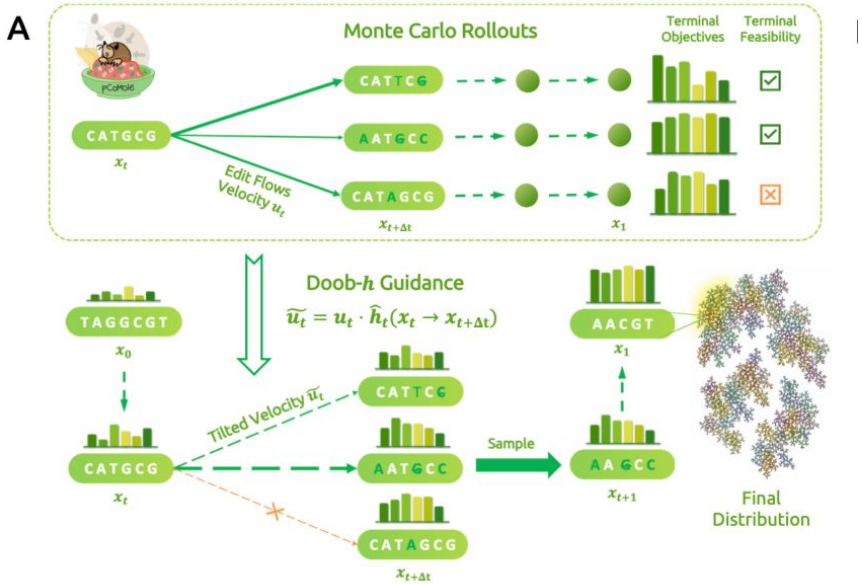
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We calculate utility from the worst objective, then exponentiate it to obtain the sampling weight.

In pCoMole, we score candidate edits by short rollouts and tilt toward the best feasible ending



Candidate edits from the base velocity, rollouts for terminal utility, then a tilted velocity.

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Shrinking a fluorescent protein to a fixed length

Method	$\lvert\lambda_{\text{exc}} - 488\rvert$ (↓)	Brightness
RayGun	22.3594	9.4207
SCISOR	20.1279	19.0333
pCoMole (UniRef$_S$ Edit Flow)	**5.8213**	43.5116
pCoMole (GFP Edit Flow)	8.4372	**49.0121**

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The sequence-model baseline misses the target excitation wavelength by 22.4 nanometres

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The second baseline misses by 20.1 nanometres and has brightness 19.0

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Chen,..,Chatterjee. *NeurIPS* (2026)

With pCoMole, the wavelength error is 5.8 nanometres and brightness is 43.5 at the same length of 213 residues

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Trained on fluorescent proteins instead, it reaches brightness 49.0 at 8.4 nanometres

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Chen,..,Chatterjee. *NeurIPS* (2026)

Let’s add guidance to our sampler

```python
import torch
import torch.nn.functional as F

def guided_step(model, prop, xt, t, h, gamma=1.0):
    oh = F.one_hot(xt, V).float().requires_grad_()
    logp = prop(oh, t).sum()                    # property on
noisy x
    g = torch.autograd.grad(logp, oh)[0]        # one backward
pass
    adv = g - (oh * g).sum(-1, keepdim=True)    # Taylor ratio
    with torch.no_grad():
        p1 = model(xt, t).softmax(-1)           # denoiser
        rate = kd(t) / (1 - k(t)).clamp(min=1e-4)
        u = rate * p1 * torch.exp(gamma * adv)  # every rate
tilted
        u.scatter_(-1, xt[..., None], 0.0)      # no
self-transition
        jump = torch.rand(xt.shape) < h * u.sum(-1)
        y = torch.distributions.Categorical(probs=u).sample()
    return torch.where(jump, y, xt)
```

We pass the one-hot noisy sequence to the property model

- We differentiate with respect to the one-hot encoding to obtain a gradient component for every letter.

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One gradient replaces a score for every destination

- We differentiate with respect to the one-hot encoding to obtain a gradient component for every letter.
- Subtract the gradient at the letter currently in place, which is the Taylor ratio.

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The unguided rates come from the denoiser, as before

- Nothing about the trained model changes.

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We multiply each rate by its guidance factor

- Nothing about the trained model changes.
- At γ = 1, we recover the exact conditional process.

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The rest of the step is unchanged

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Summary: several ways to steer one model

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For Discrete Guidance, G is a property likelihood. For pCoMole, G combines exponentiated utility with a feasibility indicator.

In MOG-DFM, we use rank and direction scores to reduce computation, at the cost of exact conditioning.

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Guidance is a Doob h-transform:

$$u_t^h(y, x) = \frac{h_t(y)}{h_t(x)}\, u_t(y, x), \qquad h_t(x) = \mathbb{E}\big[G(X_1) \mid X_t = x\big]$$

For Discrete Guidance, G is a property likelihood. For pCoMole, G combines exponentiated utility with a feasibility indicator.

In MOG-DFM, we use rank and direction scores to reduce computation, at the cost of exact conditioning.

We increase guidance strength with γ. Only γ = 1 samples the conditional distribution exactly.

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What prevents one-step discrete generation? How can we change the model to address that limitation?

Could a better pairing make the trajectories straighter and reduce the number of sampling steps?